Agriculture search and background extraction

Literature search and structured evidence extraction on AI for Social Good (AI4SG) tools in the **agriculture** sector in **Bangladesh, India, Indonesia and Thailand**.

1 Objective

Identify relevant empirical studies, evaluations, conference papers, working papers, government reports, and credible institutional reports on AI-enabled agricultural tools **that have been actually tested, deployed, implemented, or used in real farming, agricultural-extension, or agricultural-service settings** in Bangladesh, India, Indonesia, or Thailand.

Geographic scope: a source is eligible if the AI tool was tested or used in at least one of the four countries. Multi-country studies are eligible only if results, sample sizes, or implementation details for at least one of the four countries are reported separately; extract only the country-specific information.

The purpose is not to assess whether the literature is strong or weak, identify research gaps, or rank papers by causal quality.

The purpose is to extract factual information that can later be used for analysis of:

- intervention
- adoption
- effectiveness (agronomic, livelihood, environmental, programme, and agricultural-service effects)
- diagnostic and predictive performance in real settings
- implementation
- deployment
- scaling
- cost
- financing

Focus on collecting accurate and comparable information across studies.

2 Definition of AI4SG in agriculture

For this review, define AI4SG in agriculture as:

The use of AI-enabled systems to address socially important agricultural problems by improving access to agricultural information and services, the quality and timeliness of farming decisions, agricultural productivity, farmer livelihoods, food security, environmental sustainability, resilience, equity, agricultural-worker capacity, or the efficiency of agricultural-service delivery, especially where farmers, workers, communities, or institutions face resource, capacity, or access constraints.

Do not require the paper itself to use the term “AI for Social Good.”

Include an application when:

- AI is a substantive component of the tool or service;
- the application addresses an agricultural or agricultural-service problem; and

- the intended outcome concerns access, productivity, livelihoods, food security, sustainability, resilience, equity, worker capacity, affordability, or agricultural-system performance.

Include crop production, horticulture, livestock, aquaculture, extension, and agricultural value-chain services directly relevant to producers. Do not assume an application qualifies merely because it uses AI in agriculture; identify the intended social or environmental benefit from the source.

3 AI applications to include

Include, where relevant:

- AI-assisted identification of crop diseases, pests, weeds, or nutrient deficiencies from images or other field data
- point-of-use or offline AI on smartphones and portable agricultural devices
- decision support for farmers, agronomists, veterinarians, or agricultural-extension workers
- AI recommendations on planting, crop selection, irrigation, fertilizer, pesticide use, or harvest timing
- autonomous or supervised precision spraying, weeding, irrigation, and other agricultural operations
- LLM or chatbot agricultural information for farmers and extension workers
- AI-supported remote extension and agricultural consultations
- operational crop-yield prediction and weather, pest, disease, drought, or flood early-warning systems
- AI for targeting extension outreach and allocating scarce agricultural-service resources
- personalised agricultural advice and behaviour-change messaging driven by AI
- AI-based crop and agricultural-condition monitoring used operationally
- livestock and aquaculture monitoring, disease detection, feeding, and welfare support
- AI speech recognition, text-to-speech, and translation for agricultural communication in local languages
- AI-supported post-harvest grading, storage, logistics, and loss reduction
- AI-supported agricultural markets, credit, insurance, procurement, and supply-chain forecasting

Exclude ordinary digital agriculture tools where AI is not a substantive component, such as fixed-rule SMS advice, sensor dashboards, digital marketplaces, or remote consultation platforms without an AI function.

4 Type of evidence to prioritize

Prioritize sources that contain information on **AI-enabled agricultural tools that have been tested, implemented, deployed, or actually used in real agricultural settings**, including:

- randomized controlled trials
- field experiments and on-farm trials

- quasi-experimental evaluations
- diagnostic and predictive performance studies conducted in real agricultural settings
- implementation studies
- pilot and operational deployment evaluations
- scaling studies
- observational studies using actual usage data, farm records, or programme records
- government evaluations and reports
- international-organization reports
- credible programme evaluations and implementation reports
- institutional case studies documenting actual deployments
- cost, cost-effectiveness, or financing reports

Do not prioritize:

- opinion pieces and editorials
- purely conceptual papers
- general discussions or narrative reviews of AI in agriculture (use them only to identify relevant primary sources)
- attitude-only studies
- intention-to-use surveys without actual usage data
- purely technical machine-learning papers without real agricultural field testing or deployment
- studies based solely on offline validation using historical data or public datasets
- papers where AI is used only for generic convenience rather than as part of an agricultural application

Additional rules:

- Formal impact evaluations are not required. Include descriptive reports that provide useful information on actual adoption, implementation, deployment, scaling, costs, or financing.
- Distinguish research-station field testing, on-farm testing, pilot deployment, and routine operational use.
- Do not treat offline model validation as evidence of real-world testing or deployment.
- Studies using retrospective records from actual operational deployments are eligible.
- Do not prioritize studies based on research design or causal identification quality. Focus on extracting accurate, comparable factual information.

5 Search approach

Search terms

Prioritize the terms listed below and search broadly using combinations across the four categories (AI terms × agriculture terms × country and setting terms × evidence terms). These lists are a priority starting point, not an exhaustive or restrictive set. Add relevant synonyms,

technologies, applications, alternative spellings, abbreviations, programme or tool names, and local-language equivalents as needed.

AI terms

- artificial intelligence
- machine learning
- deep learning
- computer vision
- computer-aided detection or diagnosis
- agricultural decision support
- autonomous AI
- generative AI
- large language model
- chatbot or conversational agent
- risk prediction or predictive model
- natural language processing
- intelligent robotics

Agriculture terms

- agriculture or farming
- smallholder or farmer
- agricultural extension
- crop or horticulture
- plant disease, pest, weed, or nutrient deficiency
- precision agriculture
- irrigation, fertilizer, or pesticide
- yield or harvest
- livestock or aquaculture
- climate-smart agriculture or agricultural resilience
- post-harvest loss
- agricultural market, credit, or insurance

Country and setting terms

- Bangladesh
- India
- Indonesia
- Thailand
- names of states, provinces, districts, and cities
- low- and middle-income countries (then filter to the four countries)
- local-language equivalents of AI, agriculture, and the relevant application

Evidence terms

- intervention, trial, randomized, or field experiment
- real-world, on-farm, field study, or field validation
- evaluation or implementation
- adoption or usage
- deployment, scale, or scaling
- cost, cost-effectiveness, or benefit-cost
- financing or procurement
- government adoption

Sources to search

Prioritize the sources listed below. They are not the only sources to use. Search other databases, journals, proceedings, working-paper series, repositories, organizational websites, and grey-literature sources likely to contain relevant studies. Check reference lists and citing articles of relevant studies.

- Agricultural and multidisciplinary databases: AGRIS, CAB Abstracts, Scopus, Web of Science, Google Scholar, AgEcon Search
- Conference proceedings: ACM CHI, AAAI (including AI for Social Impact), IJCAI, ACM SIGKDD, NeurIPS, ICML, and relevant agricultural-engineering and agricultural-economics conferences
- Regional databases: Garba Rujukan Digital (Indonesia), Thai Journals Online (Thailand), Bangladesh Journals Online (Bangladesh), and relevant Indian agricultural journals and repositories
- Evaluation registries and repositories: AEA RCT Registry, 3ie, and relevant institutional evaluation repositories
- Grey literature: FAO, IFAD, CGIAR centres, World Bank, ADB, national agriculture ministries and research institutes, implementing NGOs, farmer organizations, and developer evidence pages

For highly relevant programmes, search the programme or tool name directly to locate additional studies or implementation reports. Where possible, connect multiple sources referring to the same AI tool or programme. Record search dates, queries, and access limitations; do not claim to have searched unavailable databases.

6. Unit of Analysis and Cross-Source Linking

Use the AI intervention or deployment as the main unit of analysis, rather than the individual paper or source.

A single AI tool may be used in multiple interventions or deployments, while multiple sources may describe the same intervention or deployment. Assign unique IDs to identify these entities and link related evidence across sources.

Assign three types of IDs:

- Tool ID: Identifies a distinct AI tool or system. Retain the same Tool ID when the tool is used across different interventions or deployments.
- Intervention ID: Identifies a distinct intervention or real-world deployment of an AI tool. Assign a new ID when the intervention design, intended application, or deployment setting differs materially.
- Source ID: Identifies each distinct paper, report, government document, webpage, or other source.

Cross-source identification and linking:

- Before assigning a new ID, check whether the AI tool or intervention has already been identified.
- Use tool/programme names, providers, implementing organizations, locations, implementation periods, target populations, and intervention designs to identify related records.
- When multiple sources describe the same intervention or deployment, retain the same Intervention ID and link all relevant Source IDs.
- When the same AI tool is used in different interventions or deployments, assign separate Intervention IDs while retaining the common Tool ID.
- Combine complementary information across related sources, including effectiveness, adoption, costs, implementation, financing, and scaling, while preserving the original source of each finding.
- Do not double-count the same intervention or outcome simply because it appears in multiple sources. Check whether different publications report the same underlying evaluation, sample, or results.
- If it is unclear whether two sources describe the same intervention, flag the relationship for verification rather than automatically merging the records.

Recording rule: Maintain one main extraction record per distinct intervention/deployment, linked to its Tool ID, Intervention ID, and all relevant Source IDs.

Background Extraction

7 Information to extract

For every relevant intervention, extract only information that is directly useful for later analysis.

A Source information

- Full citation
- Authors
- Year
- Paper/report title
- Journal, conference (and track), or institution

- DOI
- Official URL
- Trial or evaluation registration ID, if any
- Publication type

B Context

- Country
- State / province / district / city, if reported
- Country income level (World Bank classification for the study year, if the study year is not specified, use classification for the current year, and with a note)
- Rural / peri-urban / urban / mixed
- Public / private / NGO / cooperative / mixed agricultural setting
- Production or service setting (household or farm; research station; extension service; cooperative; agricultural-service provider; market or value chain; national programme)
- Agricultural subsector / crop / livestock or aquaculture species
- Target population
- Age group and sex, if relevant
- Number of farmers, households, or beneficiaries
- Number of agricultural workers, by role (extension agents, agronomists, veterinarians, farm workers)
- Number of farms, plots, facilities, or sites
- Cultivated or served area, if reported
- Farm size, tenure, irrigation or rainfed conditions, season, and other relevant sample characteristics

Distinguish people, farms, plots, animals, images, observations, and hectares. Do not treat multiple images or repeated measurements as separate beneficiaries.

C Agricultural problem

Record the main problem the AI tool is intended to address. Examples:

- delayed or incorrect detection of crop pests, diseases, or nutrient deficiencies
- shortage of agricultural-extension workers, agronomists, or veterinarians
- high workload of frontline agricultural-service workers
- limited farmer or extension-worker knowledge or training
- long travel distances and limited access to advice, inputs, or markets
- low engagement or drop-out from an agricultural programme
- low adoption of recommended farming practices
- low or unstable yields and agricultural losses
- inefficient irrigation, fertilizer, pesticide, feed, or energy use
- climate and weather risks
- post-harvest loss and inconsistent product quality
- language and literacy barriers
- financial barriers and limited access to agricultural credit or insurance

- weak pest, disease, or agricultural-risk surveillance
- inefficiency or fraud in agricultural financing and administration

Use the authors' description where possible.

D AI application

Extract:

- AI tool / programme name and version
- Provider / developer
- Regulatory status, if reported
- AI application category
- Technology type (e.g., convolutional neural network, gradient-boosting risk model, LLM with retrieval)
- Main function of the AI

Level of AI autonomy:

- assistive (human makes the final decision)
- prioritisation / triage (AI selects cases, farms, or users for further action or assessment)
- autonomous (AI initiates an operational decision or action without routine human approval)

Primary user:

- farmer / livestock keeper / aquaculture producer / beneficiary
- agricultural-extension worker
- agronomist / veterinarian / specialist
- farm worker / machinery operator
- cooperative / agricultural-service provider
- programme manager / administrator
- policymaker
- other

Application categories may include:

- detection / diagnosis
- agricultural decision support
- farmer-facing information / chatbot
- agricultural-worker information support
- risk prediction / early warning
- resource allocation / outreach targeting
- practice adoption / behaviour change
- surveillance / forecasting
- precision control / robotics
- language / translation

- markets / finance / insurance
- administration / supply chain

E Intervention design

Extract:

- What exactly was introduced?
- Who received the intervention?
- What did users do with the AI?
- What role did farmers, extension workers, or specialists play (human-in-the-loop, expert review, field confirmation)?
- Did AI replace a human assessment, prioritise cases, support an agricultural worker, advise farmers directly, operate equipment, or support administration?
- Action pathway after the AI output (e.g., field inspection, pest confirmation, input purchase, irrigation change, treatment, or extension referral)
- AI threshold or operating point used
- Reference standard / ground truth
- Duration
- Frequency
- Session length
- Intensity
- Location of use
- Control/comparison condition, if applicable

Record associated training, subsidies, free devices, input provision, or additional extension contact. Distinguish the effect of an intervention package from the isolated effect of its AI component.

F. Implementation

Extract information on the resources, conditions, and factors affecting the implementation and operation of the AI application. Record findings in three separate variables:

- Implementation Requirements: Resources, infrastructure, skills, personnel, organizational arrangements, or other conditions needed to deploy and operate the AI application.
- Implementation Barriers: Factors that hindered, delayed, limited, or increased the difficulty of implementing or operating the AI application.
- Implementation Facilitators: Factors that supported or improved the implementation and operation of the AI application.

Factors to examine (where reported):

- Training and human resources: Farmer and agricultural-worker training; staff training; technical support; supervision; monitoring; implementation staff and incentives.

- Stakeholder involvement: Farmer, extension-worker, and agricultural-specialist participation; institutional coordination.
- Human–AI interaction: Trust, resistance, over-reliance, and human oversight or overriding of AI recommendations.
- Workflow integration: Integration into farming activities, agricultural extension services, and existing organizational workflows.
- Technical infrastructure: Smartphones, sensors, cameras, drones, machinery, internet connectivity, electricity, and other equipment.
- Data and model requirements: Data availability and quality, image quality, sensor reliability, labeling, language and localization, and local agronomic relevance.
- Organizational capacity: Digital literacy, technical expertise, institutional support, quality assurance, and implementation fidelity.
- Downstream capacity: Availability of agricultural inputs, credit, labor, irrigation, and other resources needed to act on AI recommendations.
- Operational and institutional constraints: Seasonal conditions, maintenance, technical difficulties, privacy, consent, farm-data ownership, data governance, and regulatory requirements.
- Other factors: Any additional implementation requirements, barriers, or facilitators reported by the source.

Extraction Rules:

1. Classify each factor based on its role in the source, not its topic. The same factor may be a requirement, barrier, or facilitator depending on the context.
2. Extract concrete, source-supported information rather than generic statements or assumptions.
3. Distinguish observed implementation factors from anticipated challenges or recommendations for future deployment. Label anticipated or proposed factors explicitly.
4. Focus on implementation and operational delivery. Do not confuse implementation factors with adoption measures or intervention outcomes.
5. Keep entries concise while preserving relevant details and quantitative information.
6. Record "NR" for any variable not reported in the source. Do not infer that a factor was absent simply because it was not discussed.

G Deployment

Extract:

- farm / household / research station / cooperative / extension office / service centre / remote advisory setting
- agricultural production or service level where AI is used
- smartphone / tablet / laptop / desktop / basic mobile phone
- field camera / drone / satellite data / sensor / connected machinery
- IVR / voice call / SMS / messaging platform / web platform / mobile app
- cloud-based vs on-device (offline) AI
- internet requirement and offline capability

- electricity requirement
- language
- localization (local training or validation data, crop or species adaptation, locally calibrated thresholds)
- alignment with relevant agricultural guidance, if reported
- integration with existing farm, extension, market, or government information systems
- integration with government programmes, cooperatives, or agricultural insurance

H Scaling

Extract factual information on:

- initial pilot size and later deployment size
- number of farmers or beneficiaries served
- number of agricultural workers, farms, facilities, or sites
- hectares or other production units actually covered
- number of districts and states / provinces
- geographic expansion and expansion over time
- whether government adopted the tool
- integration into public extension services or a national agricultural programme
- inclusion in official guidance
- institutionalization and procurement
- regulatory approval obtained for scale-up, where relevant
- modifications made when scaling
- staffing, infrastructure, maintenance, and operational requirements at scale

Do not label something “scaled” only because authors describe it as scalable. Record actual deployment numbers where possible. Distinguish registered users, active users, cumulative reach, and future targets.

I Cost

Extract:

- total programme cost
- cost per farmer / household / farm / hectare / agricultural worker / site / user
- cost per advisory interaction, diagnosis, or agricultural operation
- marginal cost and fixed cost
- software, subscription, or per-use licence cost
- hardware/device cost
- connectivity cost
- training, staffing, technical-support, maintenance, and infrastructure costs
- downstream costs avoided (e.g., unnecessary field visits, input applications, or crop losses)
- cost-effectiveness measure (e.g., cost per additional farmer adopting a practice, cost per unit of output gained, or cost per unit of water saved)

- benefit-cost ratio, return on investment, or incremental cost-effectiveness ratio, if reported
- costing perspective (farmer, provider, programme, government, or societal)
- period over which cost is calculated
- currency and year of cost estimate

Do not combine marginal cost with total cost. Distinguish measured costs from modeled costs and separate programme costs, farmer expenditure, and estimated benefits.

J Financing

Extract:

- who paid for development, the pilot, implementation, and scale-up
- government agriculture budget / national programme funding
- agricultural insurance or financial-service funding
- cooperative or producer-organization funding
- farmer out-of-pocket payments / user fees
- donor funding and philanthropy
- technology-company grants or in-kind support
- NGO funding and development-bank funding
- private investment
- subscription, procurement, or grant funding
- public-private partnership
- results-based or outcome-based financing
- development impact bond
- other financing mechanism

When possible, distinguish research funding, pilot financing, operating financing, and scale-up financing. Do not infer a financing source unless explicitly stated.

8 Missing information

Do not guess.

If a field is unavailable, record: Not reported.

If information was searched for but could not be verified, record: Not verified.

Do not infer values from surrounding information. Preserve source references and page, table, or section locators so extracted facts can be checked.

9 Output format

Produce a structured evidence table. Use one row per intervention/deployment, not necessarily one row per paper.

For each source, produce two separate sheets. Sheet 1 (General sheet) contains the following columns; Sheet 2 (Outcome sheet) follows the Agriculture Intervention Outcome Extraction Rule. Use matching Study / Source and AI Tool labels to link the sheets, and distinguish multiple deployments in those labels where needed.

- Tool ID
- Intervention ID
- Source ID(s)
- Short Citation
- Year
- Country and State / City
- Country Income Level
- Publication Type
- Study Design
- AI Tool
- AI Function Category
- AI Function Description
- Level of AI Autonomy
- Primary User
- Agricultural Problem
- Agricultural Subsector / Crop / Species
- Production / Service Setting
- Targeted Population of Intervention
- Sample Size
- Implementing Organization
- Treatment Condition
- Control / Comparison Condition
- Farmer / Agricultural-Worker / Specialist Involvement
- Action Pathway after AI Output
- Duration
- Frequency / Intensity
- Implementation Requirements
- Implementation Barriers
- Implementation Facilitators
- Deployment Model
- Device / Infrastructure
- Language / Localization
- Deployment Stage
- Scaling Information
- Government Involvement
- Regulatory Status
- Cost Information
- Financing Source
- Financing Mechanism
- Full citation
- Official source URL

Keep each cell concise but information-rich. Where one source provides several relevant quantitative results, include the main results rather than summarizing them only as “positive” or “negative.”

10 Final output requirements

The main output should be a clean, verified evidence database that can later be used for cross-study analysis.

The goal is to collect comparable factual information on how AI agricultural tools in Bangladesh, India, Indonesia, and Thailand are designed, adopted, implemented, deployed, used, scaled, funded, and associated with agricultural, livelihood, environmental, and agricultural-system outcomes.